MILITARY STANDARD

AUTOMATED ENGINEERING
DOCUMENT PREPARATION SYSTEM

BEARINGS

TO ALL HOLDERS OF DOD-STD-35-24A(MI):

1. THE FOLLOWING PAGES OF DOD-STD-35-24A(MI) HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED.

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
3		3	15 DECEMBER 1978
4	15 DECEMBER 1978	(Reprinted without change)	
5	30 JUNE 1992	5	15 DECEMBER 1978
6	30 JUNE 1992	6	15 DECEMBER 1978
7	15 DECEMBER 1978	(Reprinted without change)	
8	30 JUNE 1992	8	15 DECEMBER 1978
9	30 JUNE 1992	9	15 DECEMBER 1978
10	30 JUNE 1992	10	15 DECEMBER 1978
11	30 JUNE 1992	11	15 DECEMBER 1978
12	30 JUNE 1992	12	15 DECEMBER 1978
13	30 JUNE 1992	13	15 DECEMBER 1978
14	15 DECEMBER 1978	(Reprinted without change)	

2. RETAIN THIS NOTICE PAGE AND INSERT BEFORE THE TABLE OF CONTENTS .

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3. Holders of DOD-STD-35-24A(MI) will verify that page changes and additions indicated above have been entered. The notice page will be retained as a check sheet. This issuance, together with appended pages is a separate publication. Each notice is to be retained by stocking points until the Military Standard i completely revised or canceled.

Custodians:
Army-MI

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Preparing activity:
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(Project EGDS-A407)

AREA EGDS

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TABLE I. TITLE INDEX

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARING, BALL, ANNULAR	112004	FF-B-171	GENERAL PURPOSE
		MIL-B-22227	MOTOR GENERATOR
		MS17107	TYPE I
		MS17108	TYPE II
		MIL-B-16129	OPEN TYPE FOR GYROSCOPES
		MIL-B-17931	QUIET OPERATION
		MIL-B-22446	HIGH GRADE
		MIL-B-81793	PRECISION
		MIL-B-83175	HIGH SPEED TURBINE ACFT
BEARING, BALL, ANNULAR, AIRFRAME, ANTIFRICTION	112001	MIL-B-7949	MINUS 65 DEG F TO 350 DEG F
		MS21428	EXTRA LIGHT DTY
		MS27640	HEAVY DUTY
		MS27641	INTMD DUTY
		MS27642	EXTRA LIGHT DTY
		MS27643	DOUBLE ROW HEAVY DUTY SELF-ALIGNING
		MS27644	DOUBLE ROW H D
		MS27645	LT AND HVY DTY, SELF-ALIGNING
		MS27646	EXTRA LIGHT DTY
		MS27647	EXTRA WIDE, DBL ROW, INTMD DUTY
		MS27648	EXTERNALLY SELF ALIGNING, EXTRA LIGHT DUTY
		MS27649	INTMD DUTY
BEARING, BALL, ANNULAR, DEEP GROOVE	112005	MIL-B-81793	PRCN FOR INSTR AND ROTATING COMPONENTS
BEARING, BALL, BELLCRANK	112002	MIL-B-6038	ANTIFRICTION
		MS20218	AIRFRAME
BEARING, BALL, DOUBLE ROW	112003	MIL-B-6039	ANTIFRICTION SEALED, ROD-END SELF ALIGNING

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARING, BALL, DOUBLE ROW	112003 (CONT)	MS21150	PRECISION, SOLID SHANK TYPE I
		MS21151	PRECISION, EXTERNAL THREAD TYPE II
		MS21152	PRECISION, HOLLOW SHANK, TYPE III
		MS21153	PRECISION INTERNAL THREAD SELF-ALIGNING TYPE IV
		MS21154	PLAIN, GROOVED OUTER RING
BEARING, BALL, DUPLEX	112031	FF-B-171	FACE TO FACE, BACK TO BACK, OR TANDEM
BEARING, BALL, THRUST	112028	FF-B-171	GENERAL PURPOSE
BEARING, RADIAL BALL	112030	MIL-B-81793	PRCN CMPNTS
BEARING, ROLLER, CYLINDRICAL	112032	FF-B-185	GENERAL PURPOSE
BEARING, ROLLER, SELF-ALIGNING, AIRFRAME, ANTIFRICTION	112022	MIL-B-8914	AIRCRAFT CONTROL SYSTEMS
		MS28912	SINGLE ROW, SEALED, TYPE I
		MS28913	DOUBLE ROW, SEALED, TYPE II
		MS28914	DOUBLE ROW, WIDE INNTER RACE SEALED TYPE III
		MS28915	DOUBLE ROW, TORQUE TUBE SEALED TYPE IV

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TABLE 1. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARINGS, JEWEL SAPPHIRE OR RUBY, SYNTHETIC	112013	MIL-B-27497	PRECISION ITEMS
		MS27041	BAR HOLE
		MS27042	OLIVE HOLE, DOUBLE CUP
		MS27043	OLIVE HOLE, RINGSTONE
		MS27045	VEE
		MS27046	CUP
		MS27047	ENDSTONE
		MS27048	OLIVE HOLE, SINGLE CUP
		MS27049	BAR HOLE, BOMBE
BEARINGS, PLAIN, ROD-END, TFE LINED, SELF-ALIGNING	112014	MIL-B-81935	SELF LUB MINUS 65 DEG F TO PLUS 325 DEG FAHRENHEIT
		MIL-B-81935/1	ROLLER, TYPE 1, EXT THD HD
		MIL-B-81935/2	ROLLER, TYPE 2, EXTERNAL THREAD HEAVY DUTY
BEARINGS, PLAIN, SELF-ALIGNING	112017	MIL-B-8942	TFE LINED
		MS21230	GROOVED OUTER RING, WIDE
		MS21231	WIDE
		MS21232	GROOVED OUTER RING, NARROW
		MS21233	NARROW
BEARINGS, PLAIN, SELF-ALIGNING, ALL METAL	112015	MIL-B-8976	AIRFRAME
		MS21154	GROOVED OUTER RING
		MS21155	PLAIN OUTER RING

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TABLE 1. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARINGS, PLAIN, SELF-ALIGNING, ALL METAL	112015 (CON'T)	MIL-B-81936	BE-CU, CRES RACE
		MIL-B-81820	SELF-LUBRICATING LOW SPEED
		MS14101	NARROW, GROOVED OUTER RING
		MS14102	WIDE, CHAMFERED OUTER RING
		MS14103	WIDE, GROOVED OUTER RING
		MS14104	NARROW, CHAM OUTER RING
BEARINGS, ROLLER	112019	MIL-B-18367	CYL, RADIAL AND THRUST, SPECIAL PRECISION
BEARINGS, ROLLER, CYLINDRICAL AND BEARINGS, ROLLER, SELF-ALIGNING	112018	FF-B-185	CYLINDRICAL AND SELF-ALIGNING
		MS19073	SINGLE ROW WITH INNER RING RET RIBS, OUTER RING RET SNAP RINGS, NON LOCATING
		MS19074	CYL OUTER RING
		MS19075	CYL INNER RING TWO RET RIBS ON OUTER RING, ONE DIR LOCATING
		MS19076	NON-LOCATING
		MS19077	OUTER RING RET RIB, DIRECTION LOCATING
		MS19078	INNER RING RET RIB ONE DIR LOCATING
		MS19079	TWO DIRECTION LOCATING

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARINGS, ROLLER, NEEDLE	112020	MIL-B-3990	AIRFRAME, ANTIFRICTION,
		MS21432	TRACK ROLLER INTEGRAL STUD, TYPE VIII
		MS24461	SINGLE ROW, HVY DTY, TYP I
		MS24462	THIN SHELL, TYPE II
		MS24463	SLFALIGNG, HVY DUTY, TYP III
		MS24464	DOUBLE ROW, HVY DTY, TYPE IV, SELF-ALIGNING
		MS24465	SINGLE ROW, HVY DTY, TRACK ROLLER, TYPE V
		MS24466	DOUBLE ROW, TYP VI, TRACK RLR
BEARINGS, ROLLER, ROD-END	112021	MIL-B-8952	ANTIFRICTION SELF ALIGNING
		MS21220	INTERNAL THREAD HVY DTY, TYPE 2
		MS21221	EXTERNAL THREAD HVY DTY, TYPE 1
		MS21223	EXTERNAL THREAD HVY DTY, TYPE 2
BEARINGS, ROLLER, TAPERED	112024	FF-B-187	GENERAL PURPOSE
		MS19081	SINGLE ROW NORMAL ANGLE
		MS19082	NORMAL ANGLE
		MS19083	STEEP ANGLE
		MS19084	STEEP ANGLE, FLANGED CUP
		MS19085	TWO ROWS, NORMAL ANGLE, ONE DOUBLE CONE, TWO SINGLE CUPS
		MS19086	TWO SINGLE CONES
		MS19087	STEEP ANGLE, ONE DOUBLE CUP

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TABLE I. TITLE INDEX (CONT)

MSE TITLE	TITLE CODE	BASE DOCUMENT NUMBER	BASE DOCUMENT COVERAGE
BEARINGS, ROLLER, TAPERED	112024 (CONT)	MS19088	NORMAL ANGLE, NON-ADJUSTABLE
		MS19089	STEEP ANGLE
BEARINGS, ROLLER, THRUST	112023	MIL-B-17380	CYLINDRICAL ROLLERS
BEARINGS, SLEEVE	112025	MIL-B-81934	PLAIN AND FLGD, SELF LUB
		MIL-B-81934/1	PLAIN TFE LINED
		MIL-B-81934/2	FLANGED, TFE LINED
		MIL-B-13506	STEEL BACKED
BEARINGS, SLEEVE, (BRONZE, PLAIN OR FLANGED)	112027	FF-B-195	TUBULAR-SHAPED
		MS13824	FLANGED, 2 X 1-25/64 INCHES
		MS13825	FLANGED, 1-3/4 X 1-13/32 INCHS
		MS13826	FLANGED, 1-5/8 X 1-3/16 INCHES
		MS13828	PLAIN, 2 X 1-7/32 INCHES
		MS13829	PLAIN, 2 X 1-1/8 INCHES
		MS13880	PLAIN, 1-3/4 X 1-7/16 INCHES
		MS13831	PLAIN, 1-5/8 X 1-3/64 INCHES
		MS13834	PLAIN, 1-9/16 X 1 INCHES
		MS13835	PLAIN, 1-7/16 X 7/8 INCHES
		MS13836	PLAIN, 1-1/2 X 3/4 INCHES
		MS13837	PLAIN, 1-7/16 X 1 INCHES
		MS13852	FLANGED, 3-3/4 X 3-3/4 INVHES
		MS13853	FLANGED, 3-3/4 X 3 INCHES
		MS13854	FLANGED, 3-3/4 X 1-21/32 INCHS

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2. REFERENCED DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this standard to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171	Bearings, Ball, Annular (General Purpose)
FF-B-185	Bearings, Roller, Cylindrical: and Bearings, Roller, Self-Aligning
FF-3195	Bearing, Sleeve, (Bronze, Plain, or Flanged)

MILITARY

MIL-B-3990	Bearing, Roller, Needle, Airframe Anti Friction
MIL-B-6038	Bearing, Ball, Bellcrank, Anti-Friction, Airframe
MIL-B-6039	Bearing, Double Row, Ball, Sealed, Rod End, Anti-Friction, Self-Aligning
MIL-B-7949	
MIL-B-3314	Bearing, Roller, Self-Aligning, Airframe, Anti-friction
MIL-B-8942	Bearings, Plain, TFE Lined, Self-Aligning
MIL-B-8948	Bearing, Plain, Rod End, TFE Lined, Self-Aligning
MIL-B-8952	Beating, Roller, Rod End, Anti Friction, Self-Aligning
MIL-B-8976	Bearing, Plain, Self-Aligning, All-Metal
MIL-B-13506	Bearing, Sleeve (Steel Backed)
MIL-B-17931	Bearing, Ball, Annular, For Quiet Operation
MIL-B-81819	Bearing, Plain, Self-Lubricating, Self-Aligning High Speed Oscillation

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MIL-13-81820	Bearing, Plain, Self-Aliging, Self-Lubricating, Low Speed Oscillation
MIL-B-81934	Bearing, Sleeve, Plain and Flanged, Self-Lubricating
MIL-B-81935	Bearings, Plain, Rod End, Self-Aligning, Self-Lubricating
MIL-B-81936	Bearings, Self-Aligning, (BeCu, Cres Race)

STANDARDS

MILITARY

DOD-STD-35	Automted Engineering Document preparation System
MIL-STD-130	Identification Marking of Military Property
MIL-STD-206	Friction Torque Testing for Instrument Ball Bearings. Parts 1 and 2.
MIL-STD-810	Environmental Test Methods
MIL-STD-1523	Age Control of Age-Sensitive Elastomeric Materials

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ASTMD256	Impact Resistance of Plastics and Electrical Insulating Materials, Methods of Test For
ASTMM38	Tensile Properties of Plastics, Test for
ASTMD695	Compressive Properties of Rigid Plastics, Test For
ASTMD732	Shear Strength of Plastics, Standard Method of Test For
ASTMD790	Flexural Properties of Plastics, Test For

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